

6.2 Lin. op. f na prostora V :

$$V \cong \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{R}^4$$

$$\downarrow \quad \downarrow$$

$$\vec{v}_1 \quad \vec{v}_2$$

splňuje: $f\left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}$

$$\downarrow$$

$$\vec{v}_1 + 2\vec{v}_2$$

$$f\left(\begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$$

$$\downarrow$$

$$2\vec{v}_1 - 2\vec{v}_2$$

Spočítaj $f^h\left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix}\right) \rightarrow \vec{v}_1 + 3\vec{v}_2$

$B = (\vec{v}_1, \vec{v}_2)$

$$[f]_B = \left(\left[\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \right]_B \mid \left[\begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \right]_B \right) = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

$\hookrightarrow$ keďže $\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \in V$

$\hookrightarrow$ Viem ju diag?
Skús si to.

$\det([f]_B - \lambda I_2) = \det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = (\lambda+3)(\lambda-2)$

$\lambda_1 = -3 \quad M_{-3} = \ker \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\}$

$\lambda_2 = 2 \quad M_2 = \ker \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} = \text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$

$\downarrow$
viem $[f]_B$ diagonalizovať, pokiaľ ju ľahko urobím

$C = \left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right), [f]_B^C = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$

$$[f^h]_B = [id]_{U_2}^C [f^h]_B^C ([id]_{U_2}^C)^{-1}$$

$$= \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} (-3)^h & 0 \\ 0 & 2^h \end{pmatrix} \frac{1}{5} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} (-3)^h + 4 \cdot 2^h & -2(-3)^h + 2 \cdot 2^h \\ -2(-3)^h + 2 \cdot 2^h & 4(-3)^h + 2^h \end{pmatrix} \cdot \frac{1}{5}$$

nech $\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \vec{u} \Rightarrow [\vec{u}]_B = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

$$[f^h]_B \cdot [\vec{u}]_B = \begin{pmatrix} - & 1 & - \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} (-3)^h + 4 \cdot 2^h & -2(-3)^h + 2 \cdot 2^h \\ -2(-3)^h + 2 \cdot 2^h & 4(-3)^h + 2^h \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

označ a

označ b

$$\Rightarrow [f^h \cdot \vec{u}]_B$$

$\Rightarrow f^h \cdot \vec{u} = \frac{1}{5} a \cdot \vec{v}_1 + \frac{1}{5} b \cdot \vec{v}_2$